

Physics A (H156, H556)

Exploring - Point and spherical charges KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is 53.
- The total number of marks may take into account some 'either/or' question choices.

1 This question is about lightning.

Sheet lightning occurs when there is an electrical discharge between the upper and lower regions of a thunder cloud.

The upper regions are positive and the lower regions are negative.

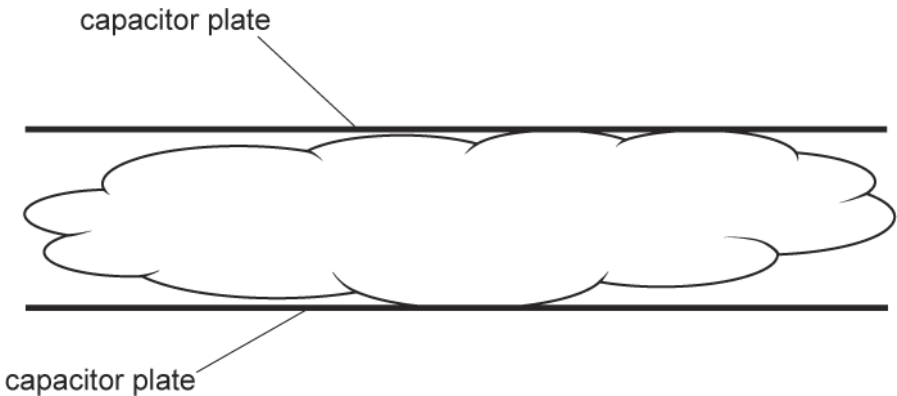
The thunder cloud can be modelled as an ideal parallel plate capacitor with circular horizontal plates.

The data for the capacitor comes from the cloud.

Diameter of cloud	24 km
Distance between upper and lower regions	3.2 km
Electric field strength between the regions	$4.0 \times 10^5 \text{ V m}^{-1}$

i. The diagram shows the plates of the model capacitor superimposed on the cloud.

Draw on the diagram to show the electric field lines between capacitor plates.



[2]

ii. Suggest why the actual electric field lines of the cloud would differ from what you have drawn.

[1]

iii. Show that the potential difference (p.d.) V between the plates is about $1 \times 10^9 \text{ V}$.

[1]

- iv. Calculate the capacitance C of the model capacitor.

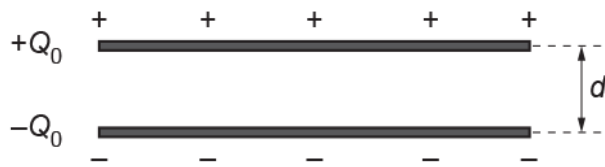
Assume the permittivity of the material of the cloud is the same as the permittivity of free space.

$$C = \dots\dots\dots \text{ F [2]}$$

- v. Calculate the magnitude of the charge Q on one of the plates of the model capacitor.

$$Q = \dots\dots\dots \text{ C [2]}$$

The diagram below shows a simple capacitor.



The capacitor consists of two horizontal metal plates in a vacuum. The magnitude of the charge on each plate is Q_0 . The potential difference (p.d.) between the plates is V_0 . The capacitor plates have capacitance C_0 . The separation between the plates is d . The energy stored by the capacitor is E_0 .

The top plate is moved vertically upwards. The new separation between the plates is $2d$.

The charge on each plate remains the **same**.

The energy stored by the capacitor **increases**.

i. Determine the new:

1 capacitance in terms of C_0

capacitance = C_0 [1]

2 p.d. between the plates in terms of V_0

p.d. = V_0 [1]

3 energy stored in terms of E_0 .

energy = E_0 [1]

ii. Explain, in terms of forces between the plates, why the energy stored increases.

----- [1]

3(a) The diagram below shows two parallel plates, E and C, in an evacuated glass tube.

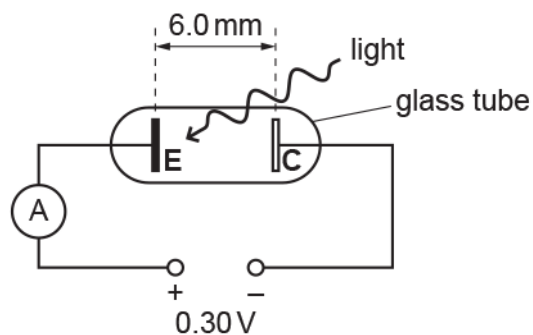


Plate E is made from potassium, which is sensitive to light. Plate C is not sensitive to light.

The separation between the plates is 6.0 mm and the potential difference between the plates is 0.30 V.

Light of frequency 6.3×10^{14} Hz is incident on plate E. The photoelectrons emitted from this plate have **maximum** kinetic energy 0.30 eV (4.8×10^{-20} J). The photoelectrons are repelled by the negative plate C. The ammeter reading is zero because these photoelectrons reach plate C with zero kinetic energy.

This question is about a photoelectron emitted perpendicular to plate E and with an initial kinetic energy of 4.8×10^{-20} J.

- i. Show that the magnitude of deceleration of this photoelectron is $8.8 \times 10^{12} \text{ ms}^{-2}$.

[3]

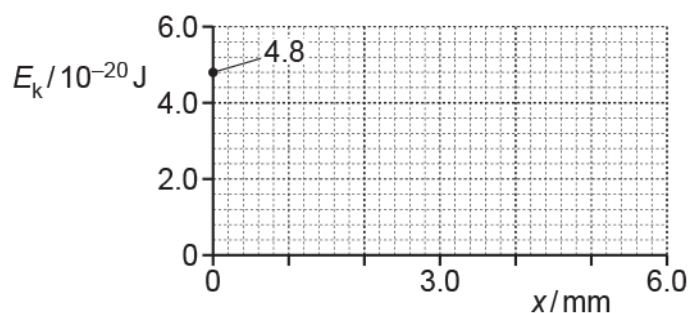
- ii. Show that the initial speed of the photoelectron is about $3 \times 10^5 \text{ ms}^{-1}$.

[2]

- iii. Calculate the time t taken by the photoelectron to travel from plate E to plate C.

$t = \dots\dots\dots$ s [2]

iv. Using the axes shown below, sketch a graph of kinetic energy E_k against distance x from plate E.



[2]

(b) Explain, in terms of photons, what happens to the ammeter reading when light of frequency greater than 6.3×10^{14} Hz is now incident on plate E.

----- [2]

4 Fig. 24 shows two horizontal metal plates in a vacuum.

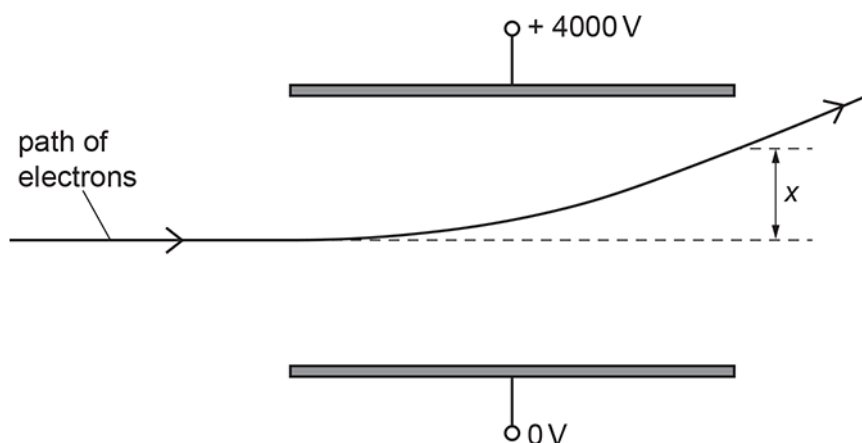


Fig. 24

The diagram is **not** drawn to scale.

Electrons travelling horizontally enter the space between the charged plates and are deflected vertically.

The potential difference between the plates is 4000 V.

The distance between the plates is 0.08 m.

The initial speed of the electrons is $6.0 \times 10^7 \text{ m s}^{-1}$.

The vertical deflection of the electrons at the far end of the plates is x .

- i. Show that the vertical acceleration a of an electron between the plates is $8.8 \times 10^{15} \text{ m s}^{-2}$.

[3]

- ii. The length of each plate is 0.12 m.

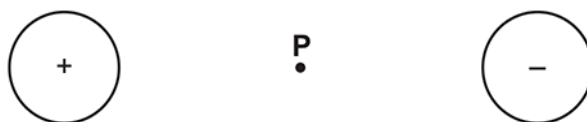
Show that the time t taken by the electron to travel this length is $2.0 \times 10^{-9} \text{ s}$.

[1]

- iii. Calculate the vertical deflection x of the electron.

$x = \dots\dots\dots \text{ m}$ [2]

5 The diagram below shows two oppositely charged spheres.



The magnitude of the charge on each sphere is the same.

The point **P** is on the line joining the centres of the spheres and is the same distance from the centre of each sphere.

Which statement is correct?

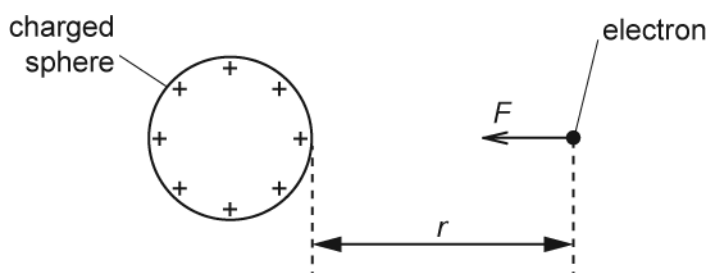
- A A negatively charged particle at **P** will move to the right.
- B The direction of the electric field at **P** is to the left.
- C The electric potential at **P** is zero.
- D The magnitude of the electric field strength at **P** is zero.

Your answer

☐

[1]

- 6 An electron is released at a distance r from the surface of a positively charged sphere. It is attracted towards the centre of the sphere and moves until it touches the surface.



Which of the following statements is/are correct?

- 1 The area under the F against r graph is equal to work done on the electron.
- 2 The electric field strength E at distance r is equal to $\frac{F}{1.6 \times 10^{-19}}$.
- 3 The work done on the electron is equal to $F \times r$.

- A Only 1
 B Only 1 and 2
 C Only 1 and 3
 D 1, 2 and 3

Your answer

[1]

7(a)

Fig. 20.1 shows a positively charged metal sphere and a negatively charged metal plate.

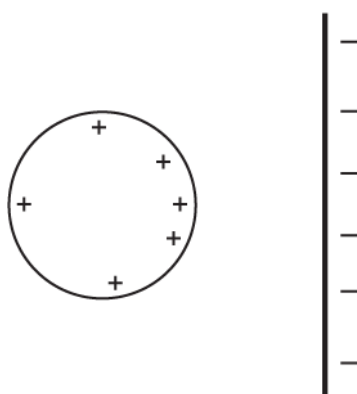


Fig. 20.1

On Fig. 20.1, draw a minimum of **five** electric field lines to show the field pattern between the sphere and the plate.

[2]

(b) Define *electric potential* at a point in space.

[1]

(c) A metal sphere is given a positive charge by connecting its surface briefly to the positive terminal of a power supply. The electric potential at the surface of the sphere is + 5.0 kV. The sphere has radius 1.5 cm.

i. Show that the charge Q on the surface of the sphere is 8.3×10^{-9} C.

[2]

ii. Fig. 20.2 shows the charged sphere from (i) suspended from a nylon thread and placed between two oppositely charged vertical plates.

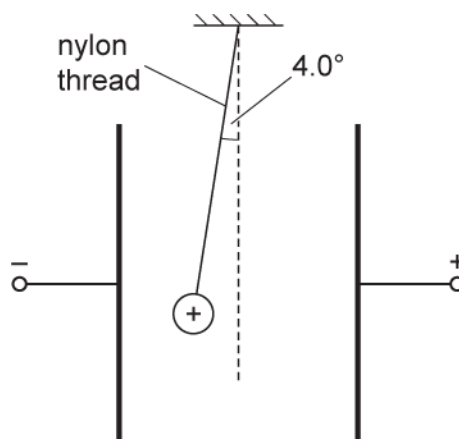


Fig. 20.2 (not to scale)

The weight of the sphere is 1.7×10^{-2} N. The string makes an angle of 4.0° with the vertical.

1. Show that the electric force on the charged sphere is 1.2×10^{-3} N.

[1]

2. Calculate the uniform electric field strength E between the parallel plates.

$$E = \text{-----} \text{ N C}^{-1} \text{ [2]}$$

8(a) Fig. 21.1 shows two oppositely charged ions to the left of a point X.



Fig. 21.1

The separation between the centres of the ions is 3.0×10^{-10} m. Each ion has charge of magnitude 1.6×10^{-19} C.

- i. Explain why the direction of the **resultant** electric field strength at point X is to the left.

----- [2]

- ii. Calculate the minimum energy in eV required to completely separate the ions.

energy = ----- eV [3]

(b) A **proton** travels from point **P** to point **Q** in a uniform electric field as shown in Fig. 21.2.

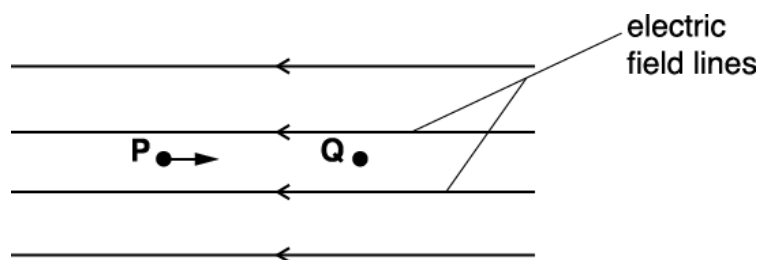


Fig. 21.2

The velocity of the proton at **P** is $7.2 \times 10^6 \text{ m s}^{-1}$ and the velocity at **Q** is $2.4 \times 10^6 \text{ m s}^{-1}$. The distance between **P** and **Q** is 1.2 cm.

Calculate

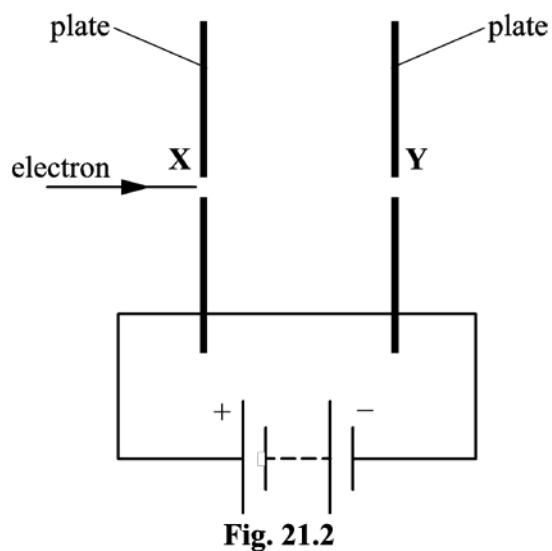
- i. the magnitude of the deceleration of the proton

deceleration = _____ m s^{-2} [2]

- ii. the electric field strength E .

$E =$ _____ N C^{-1} [2]

9 Fig. 21.2 shows two parallel vertical metal plates connected to a battery.



The plates are placed in a vacuum and have a separation of 1.2 cm. The uniform electric field strength between the plates is 1500 V m^{-1} . An electron travels through holes X and Y in the plates. The electron has a horizontal velocity of $5.0 \times 10^6 \text{ m s}^{-1}$ when it enters hole X.

- i. Draw five lines on Fig. 21.2 to represent the electric field between the parallel plates.

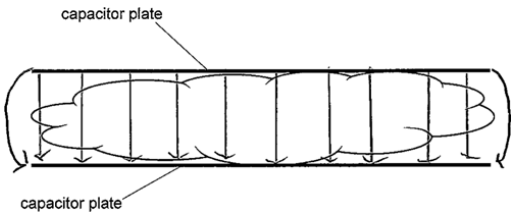
[2]

- ii. Calculate the final speed of the electron as it leaves hole Y.

speed = _____ m s^{-1} [3]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1		i	At least 4 equidistant parallel vertical (straight) lines With arrows pointing downwards	B1 B1	By eye Ignore field lines outside of plates At least one line must touch top and bottom plate At least one arrow, and all arrows given must be correct <u>Examiner's Comments</u> Most candidates were able to score at least 1 mark on this question. The most common loss of marks was an uneven spacing between the lines. Some candidates (helpfully) wrote on the diagram that the spacing was equal. As it is apparent that most candidates know the spacing should be regular, it would be best to do this with a ruler rather than leave it judgement. Similarly, the lines should be vertical and straight, again best done with a ruler. Exemplar 3  Exemplar 3 only scored the second mark, as the spacing is unequal. At first glance it may look like the spacing is regular and this is possible what the candidate meant to do, but those at the right hand side (ignoring those outside of the plates) are definitely not equally spaced by eye. It would have been far better for the candidate to use a ruler to make the spacing equal.

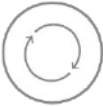
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	The top and the bottom of the clouds will not be parallel / will be uneven	B1	Allow uneven charge distribution / charge density Allow the field is non-uniform Allow the cloud is a non-uniform shape Ignore cloud has non uniform density <u>Examiner's Comments</u> Although there were many ways to obtain this mark, only around half of candidates gave a suitable response. Many answers were vague such as 'the cloud is not even' or 'the cloud contains water vapour' which really needs more clarity. The best response is based on the distribution of charges, but alternatives relating to uniformity of a variety of factors are perfectly correct.
		iii	$(V = Ed) = 4.0 \times 10^5 \times 3.2 \times 10^3 = 1.3 \times 10^9 \text{ (V)}$	B1	Values of E, d with correct powers of 10 and correct evaluation must be seen to at least 2sf (1.28) Ignore unit <u>Examiner's Comments</u> Again, a 'show that' question needs the calculation to be clear. Here it will involve the multiplication of two quantities using the correct powers. For example, 32k is not a suitable alternative to 32×10^3. The value calculated is not the same as the 'show that' value so needs to be given to more significant figures (at least 2) to prove the calculation was carried out. The significant majority of candidates were able to correctly score this mark.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iv	$C = \left(\frac{\epsilon_0 A}{d} \right) = \frac{8.85 \times 10^{-12} \times \pi \times (12 \times 10^3)^2}{3.2 \times 10^3}$ $C = 1.3 \times 10^{-6} \text{ (F)}$	C1 A1	All values (including ϵ_0) substituted correctly Correct evaluation to at least 2sf (1.25×10^{-6}) Note: use of $C = 4\pi\epsilon R$ (leading to 1.3×10^{-6} is XP) <u>Examiner's Comments</u> Most candidates were able to select the correct formula and make an attempt at a substitution. The main error came from an incorrect calculation of the area, either by using an incorrect method for the circle or incorrectly calculating it using $24\text{km} \times 3.2\text{km}$. Candidates are to be reminded that if the working is correct and the calculation wrong, then marks may still be given. Without this working, there is likely to be no credit. Several candidates used the formula for charge on a sphere which gave the same answer (to 2sf) but is a physics error so scores no marks.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		v	$Q (= CV) = 1.25 \times 10^{-6} \times 1.28 \times 10^9$ $Q = 1600 \text{ (C)}$	C1 A1	Ecf from (a)(iii) and (a)(iv) Correct evaluation to at least 2sf Allow 2sf answer of 1700 C for 2sf values used in calculation Allow answer of 1250C for p.d. value of 1×10^9 Examiner's Comments This response could produce a wide variety of correct answers depending on the rounding of the numbers used, although most candidates used at least 2sf for their values, which is recommended. A noticeable number of candidates correctly evaluated the change then divided it by 2, presumably due to it asking for the charge on one of the plates.  Assessment for learning Error carried forward. In general, error carried forward can only be applied when the working is clearly seen and the error value is correctly used. In this question, there are two values to be used, each of which may have been incorrectly calculated. It is therefore vital that the calculation is seen, so that the credit can be given. It is, of course, good practice to always show working rather than when just an error carried forward could be applied.
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
2		i	1 0.5 (C_0) 2 2 (V_0) 3 2 (E_0)	B1 B1 B1	Allow $\frac{1}{2}$ Ignore working No ecf Ignore working No ecf <u>Examiner's Comments</u> Around two thirds of candidates were able to score all of these marks. Most showed some limited (but helpful) working, such as writing the equation for the parallel plate capacitor and $C = Q/V$, to assist them in appreciating how each of the factors change. For this question, there is a quite large amount of introductory text and the bold text is there as a supportive guide. The most common incorrect responses were a simple reversal of the correct responses.
		ii	<u>Work</u> done against <u>attractive</u> forces	B1	Allow WD for work done <u>Examiner's Comments</u> This proved to be a challenging question and only the higher end candidates were able to give a clear and correct response. The question stated, "in terms of forces" and most candidates did not explain the idea of attraction between the plates. Common incorrect responses included using $E = \frac{1}{2} QV$ or using $W = F \times d$ as a starting point.
			Total	4	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
3	a	i	$a = \frac{vQ}{dm} \text{ OR } a = \frac{EQ}{m} \text{ OR } KE = \frac{1}{2} mv^2 \text{ and } v^2 = u^2 + 2as$ $\text{OR } KE = F \times d \text{ and } F = m \times a$ $a = \frac{0.30 \times 1.6 \times 10^{-19}}{6.0 \times 10^{-3} \times 9.11 \times 10^{-31}} \quad / \quad a = \frac{50 \times 1.6 \times 10^{-19}}{9.11 \times 10^{-31}} \quad /$ (Use of $KE = \frac{1}{2} mv^2$) = $4.8 \times 10^{-20} = \frac{1}{2} \times 9.11 \times 10^{-31} \times v^2$ and (use of $v^2 = u^2 + 2as$) = $v^2 = (1.05 \times 10^{11}) = 2 \times a \times 6 \times 10^{-3} (\pm 0^2)$ (Use of $KE = F \times d$) = $4.8 \times 10^{-20} = F \times 6 \times 10^{-3}$ and (use of $F = m \times a$) $F = (8.0 \times 10^{-18}) = 9.11 \times 10^{-31} \times a$ $a = 8.78 \dots \times 10^{12} \text{ (ms}^{-2}\text{)}$	C1 M1 A1	Allow u and v interchangeably throughout Allow calculation of $E = (0.30 / 6 \times 10^{-3}) = 50 \text{ (V m}^{-1}\text{)}$ or $v = 3.2 \times 10^5 \text{ (ms}^{-1}\text{)}$ or $v^2 = 1.05 \times 10^{11} \text{ (ms}^{-1}\text{)}^2$ or $F = 8.0 \times 10^{-18} \text{ (N)}$ for C1 mark Substitution mark – in any arrangement. Expect full substitutions including numerical value of m_e if appropriate Method 1: direct calculation of a Method 2: using $KE = \frac{1}{2} mv^2$ and $v^2 = u^2 + 2as$ Method 3: using $KE = F \times d$ and $F = m \times a$ Note must be more than 2 SF (not paper SF penalty) Ignore negative sign Examiner's Comments There were many different routes to showing the acceleration, and marks were given for each method or part method. No one method was seen significantly more than others, and some candidates used a variety of pathways to come to their answer. The main principle in the question (and the subsequent one) where the candidate is being asked to “show that” a given value is correct is that the examiner must be convinced that the candidate has clearly demonstrated that they have carried out the calculation and evaluated it on their calculator. The instructions which examiners used was: first marking point for providing one (or two) equations that would lead to the solution, or calculation of an intermediate value; second marking point for a full substitution into one or more equations; third marking point for using this full substitution to produce an answer to more sf than given in the question. As the second marking point was deemed to be an M mark, the full substitution needed to be seen to gain the A mark. A small number of (often higher end)

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			candidates did not show the full substitution, often missing out the value of m_e in their calculation, and another common error was to not show the extra significant figure. Over half of the candidates were able to achieve full marks on this question and it generally discriminated well. Assessment for learning When a question asks a candidate to “show that” a given value is correct, the following two points should be considered: • Each stage of the calculation should be clearly shown. Preferably setting out any equation first, and then showing a full substitution of all values into that equation • If the value calculated by the candidate would correctly round to the given value, then the candidate should show their calculated value to at least one more significant figure than the given value. Both of these are evidence that the complete calculation has taken place and that the candidate has not somehow artificially generated the required value. This advice should be viewed as “best practice” rather than a rigid set of rules. Reverse arguments are often possible where a candidate can work backwards from their given value, however this is not the advised approach.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	(Use of $KE = \frac{1}{2} mv^2$) = $4.8 \times 10^{-20} = \frac{1}{2} \times m \times v^2$ OR ($u^2 = v^2 - 2as$) = $0^2 - [2 \times (-) 8.8 \times 10^{12} \times s]$ Full substitution leading to $v = 3.2 \dots \times 10^5$ (ms^{-1})	C1 A1	Allow u and v interchangeably Numerical value of m_e must be used if using <i>KE</i> method Note must be more than 1 SF (not paper SF penalty) Note 3.25 is acceptable for A1, but not 3.3 <u>Examiner's Comments</u> The vast majority of candidates were able to clearly show that the speed of the photoelectron could be calculated as $3.2 \times 10^5 \text{ ms}^{-1}$, most often through substitution into the kinetic energy formula. As in Question 19 (b) (i), it is important to show all variables and constants used in the equation for full marks and to give the answer to at least one more d.p. than given in the question, to show the calculation has taken place. An alternative solution using an equation of motion and the acceleration given (or calculated) in Question 19 (b) (i) would yield the same result.
		iii	$t = \frac{3.2 \times 10^5}{8.8 \times 10^{12}}$ $t = 3.6 \times 10^{-8} \text{ (s)}$	C1 A1	Allow correct full substitution into any suvat equation Allow 3×10^5 for v Ignore signs of substituted values Expect values between 3.4×10^{-8} and $3.8 \times 10^{-8} \text{ (s)}$ No ecf from (b)(i) or (b)(ii) <u>Examiner's Comments</u> Only around half of the candidates were able to obtain answers within the required range. Candidates used a variety of rounded or none-rounded values from prior calculations, so a generous range of responses was given to allow for this. A common error among less successful responses was to simply use speed = distance/time usually leading to $2.0 \times 10^{-8} \text{ s}$. Those using $s = ut + \frac{1}{2} at^2$ often encountered problems in solving the equation as it could lead to imaginary roots and tended to produce solutions way outside of the accepted values. However, marks could be given for setting up the calculation correctly.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iv	Any line with negative slope starting from 0, 4.8×10^{-20} A straight line finishing at 6.0,0	M1 A1	$\frac{1}{2}$ square tolerance. ALLOW a curve with negative gradient. ALLOW a region of zero gradient, but not whole line $\frac{1}{2}$ square tolerance on axis intercepted <u>Examiner's Comments</u> Nearly all candidates appreciated that this line would start at 4.8×10^{-20} J and decrease to zero at 6.0 mm. However, the vast majority drew a curved line of decreasing gradient. This may well have come from a confusion from $KE \propto v^2$ and attempting to draw a parabola.
	b		Energy of photon increases (max) kinetic energy / speed (of electrons) increases / (some) electrons (now) reach C and there is a current or reading (on ammeter)	B1 B1	Do not allow increased <i>kinetic</i> energy of photons Do not allow explanation in terms of increased number of emitted electrons (per second) Allow photoelectrons for electrons <u>Examiner's Comments</u> There were several misconceptions in candidates' responses to this question. Many candidates did not appreciate that the increased frequency would result in electrons of greater KE and thought that it was the increased energy of the photons crossing the 6.0mm gap that caused an ammeter reading. A significant number of candidates also described increasing frequency causing an increase in <i>kinetic</i> energy of photons, and some also linked the increasing frequency to a greater number of photons or photoelectrons.
			Total	11	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4		i	$(E =) \frac{4000}{0.080}$ $(F =) \frac{4000}{0.080} \times 1.6 \times 10^{-19}$ $(a =) \frac{8.0 \times 10^{-15}}{9.11 \times 10^{-31}} \text{ or } 8.78 \times 10^{15}$ $a = 8.8 \times 10^{15}$	C1 C1 C1 A0	$E = 5.0 \times 10^4 \text{ (V m}^{-1}\text{)}$ $F = 8.0 \times 10^{-15} \text{ (N)}$ Allow this mark if the working is shown. If only value is given, then the answer must be 3SF or more <u>Examiner's Comments</u> This question asks for a calculation to show the value of the vertical acceleration in an electric field. The magnitude of the electric field strength first needs to be calculated, followed by the acceleration from Newton's second law. Candidates are reminded that a show question needs to be answered in detail and that each stage should be clear. Roughly equal numbers of candidates scored full marks or zero on this question.
		ii	$(t =) \frac{0.12}{6.0 \times 10^7}$ $(t = 2.0 \times 10^{-9} \text{ s})$	M1 A0	<u>Examiner's Comments</u> As with the previous question, there is the need to make sure that the calculation leading to the given answer is clearly set out.
		iii	$(x =) \frac{1}{2} \times 8.78 \times 10^{15} \times (2.0 \times 10^{-9})^2$ $x = 1.8 \times 10^{-2} \text{ (m)}$	C1 A1	Allow $a = 8.8 \times 10^{15}$ <u>Examiner's Comments</u> Most candidates appreciated the need to use an equation of motion in their solution, but a significant number of candidates used an initial horizontal velocity in the expression, leading to an incorrect answer. There were also an unusually large number who gave no response. Candidates should appreciate that if they have been given show questions, then it is likely that these values will be used in alter questions.  Misconception Many candidates included an initial vertical velocity – it may be helpful to think of this process as analogous to that of projectile motion.
			Total	6	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
5			C	1	<u>Examiner's Comments</u> The correct response is C. Candidates often find the electric field questions challenging and this was again the case as this question was correctly answered by only one third of the candidates. Many candidates drew arrows on the diagram to assist them. Response D was the most common distractor; linking this to gravitational fields would produce a zero field strength at P which is likely to be the reason.
			Total	1	
6			B	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
7	a		Correct pattern	B1	Note: At least five field lines must be drawn and of these, two must be perpendicular (by eye) to the surface of the sphere and plate
			Correct direction of the field	B1	Note: This may be shown on just one line Examiner's Comment Most candidates drew decent field patterns and showed the correct direction of the electric field. It is difficult to draw curved field lines, but those who were careful and had the field lines perpendicular at both the surface of the sphere and the metal plate were rewarded.
	b		(Electric potential) is the <u>work</u> done per (unit) charge in bringing a <u>positive</u> charge from infinity (to the point).	B1	Allow: <u>work</u> done / <u>energy</u> required to bring a unit <u>positive</u> charge from infinity (to the point) Examiner's Comment This was not well-answered; the modal mark was zero. Definition for electric potential lacked precision and often made no reference to a 'unit positive charge' or 'per unit positive charge'. At times, other quantities such as electric field strength and gravitational field strength were being defined. This was a missed opportunity -definitions just need to be learnt.
	c	i	$V = Q/4\pi\epsilon_0 r$ (Allow any subject) $Q = 4\pi \times 8.85 \times 10^{-12} \times 0.015 \times 5000$ $Q = 8.3(4) \times 10^{-9} \text{ (C)}$	C1 C1 A0	Note using $E = V/d$ with $E = Q/4\pi\epsilon_0 r^2$ is wrong physics and hence scores zero Note if the value of ϵ_0 is not given here, it could be implied in the correct 3sf answer Allow any subject here if the answer is given to more than 2sf Allow the use of $1/4\pi\epsilon_0 = 9 \times 10^9$ Examiner's Comment By contrast to the last question, the answers here were perfect. Correct values were substituted into the equation for electric potential to show that the charge was that stated in the question. In a 'show' question, always give the final answer to more significant figures than the required answer. It was good to see many scripts with the final answer written as $8.34 \times 10^{-9} \text{ C}$.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	1 (electric force =) $1.7 \times 10^{-2} \times \tan 4.0$ (Allow any subject) (electric force = 1.19×10^{-3} N) 2 $E = 1.2 \times 10^{-3} / 8.3(4) \times 10^{-9}$ $E = 1.4 \times 10^5$ (N C⁻¹)	M1 (A0) C1 A1	Not $1.7 \times 10^{-2} \sin 4$ or $1.7 \times 10^{-2} \cos 86$ Allow $1.7 \times 10^{-2} \times \sin 4 / \cos 4$ Allow 2 marks for 1.45×10^5 (N C⁻¹), 8.3×10^{-9} used Allow 2 marks for 1.43×10^5 (N C⁻¹), 1.19×10^{-3} (N) used Examiner's Comment This was a good discriminator with high-scoring candidates either using triangle of forces, or resolution of forces, to determine the electric force on the sphere. The value of the force was given so that it could be used to answer the next question. More than half of the candidates correctly calculated the electric field strength using the information provided in (c)(i) and (c)(ii)1. Some candidates used the elementary charge rather than the value from (c)(i) to calculate the field strength; this gave an incorrect answer of 7.5×10^{15} N C⁻¹.
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
8	a	i	The direction of the electric field due to the negative charge is to the left and right for the positive charge.	B1	
		i	The magnitude of the electric field strength due to the positive charge is smaller than that for the negative charge (because of greater distance). (Hence the resultant electric field strength is to the left.)	B1	
		ii	$\text{energy} = \frac{Qq}{4\pi\epsilon_0 r} = \frac{(1.60 \times 10^{-19})^2}{4\pi\epsilon_0 \times 3.0 \times 10^{-10}}$	C1	
		ii	energy = 7.67(2) × 10 ⁻¹⁹ (J)	C1	
		ii	energy = 4.8 (eV)	A1	
	b	i	$(v^2 = u^2 + 2as)$ $(2.4 \times 10^6)^2 = (7.2 \times 10^6)^2 + 2 \times a \times 1.2 \times 10^{-2}$	C1	Allow other correct methods
		i	$a = (-) 1.9 \times 10^{15} \text{ (m s}^{-2}\text{)}$	A1	Allow 1 mark for 1.9 × 10 ¹³ ; distance left in cm Note answer to 3 s.f. is 1.92 × 10 ¹⁵ (m s ⁻²) Ignore sign
		ii	$E = F/Q$ and $F = ma$	C1	Possible ECF from (i)
		ii	$E = \frac{1.67 \times 10^{-27} \times 1.92 \times 10^{15}}{1.60 \times 10^{-19}}$	C1	
		ii	$E = 2.0 \times 10^7 \text{ (N C}^{-1}\text{)}$	A1	
			Total	9	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
9		i	Parallel and equidistant field lines.	B1	Note: Field lines must be right angle to the plates.
		i	Field direction is correct (from left to right).	B1	
		ii	work done = $1500 \times 1.6 \times 10^{-19} \times 1.2 \times 10^{-2} = 2.88 \times 10^{-18}$ (J)	C1	Correct use of: final KE = initial KE – work done.
		ii	$\frac{1}{2} \times 9.11 \times 10^{-31} \times v^2 = \frac{1}{2} \times 9.11 \times 10^{-31} \times (5.0 \times 10^6)^2 - 2.88 \times 10^{-18}$	C1	
		ii	speed = 4.3×10^6 (m s ⁻¹)	A1	
			Total	5	